

Part A

Database Creation

CREATE DATABASE CollegeDB;

USE CollegeDB;

Create Students Table

```
CREATE TABLE Students (  
    StudentID INT PRIMARY KEY,  
    FirstName VARCHAR(50) NOT NULL,  
    LastName VARCHAR(50) NOT NULL,  
    Email VARCHAR(100) UNIQUE  
);
```

Create Courses Table

```
CREATE TABLE Courses (  
    CourseID INT PRIMARY KEY,  
    CourseName VARCHAR(100) NOT NULL  
);
```

Create Enrollments Table

```
CREATE TABLE Enrollments (  
    EnrollmentID INT PRIMARY KEY,  
    StudentID INT,  
    CourseID INT,  
    FOREIGN KEY (StudentID) REFERENCES Students(StudentID),  
    FOREIGN KEY (CourseID) REFERENCES Courses(CourseID)  
);
```

Part B

Insert Student Records

INSERT INTO Students VALUES

(1, 'Fatima', 'Ali', 'Fatima@gmail.com'),
(2, 'Shacifi', 'Shuceyb', 'Shacifi@gmail.com'),
(3, 'Anas', 'Hassan', 'Anas@gmail.com'),
(4, 'Marwan', 'Ali', 'Marwan@gmail.com'),
(5, 'Abdi Rahman', 'Omar', 'ali@gmail.com');

Insert Course Records

INSERT INTO Courses VALUES

(101, 'Database Systems'),
(102, 'Programming Fundamentals'),
(103, 'Web Development'),
(104, 'Networking'),
(105, 'Data Structures');

Insert Enrollment Records

INSERT INTO Enrollments VALUES

(1, 1, 101),
(2, 2, 102),
(3, 3, 103),
(4, 4, 104),
(5, 5, 105);

Part C

1. Display all students

```
SELECT * FROM Students;
```

2. Display all courses

```
SELECT * FROM Courses;
```

3. Display students ordered by name

```
SELECT * FROM Students  
  
ORDER BY FirstName ASC;
```

4. Find a student using StudentID

```
SELECT * FROM Students  
  
WHERE StudentID = 1;
```

5. Count total students

```
SELECT COUNT(*) AS TotalStudents FROM Students;
```

6. Display students whose names start with A

```
SELECT * FROM Students  
  
WHERE FirstName LIKE 'A%';
```

Part D

1. Display student names with their courses

```
SELECT Students.FirstName,  
       Students.LastName,  
       Courses.CourseName  
  
FROM Enrollments  
  
INNER JOIN Students  
  
ON Enrollments.StudentID = Students.StudentID  
  
INNER JOIN Courses  
  
ON Enrollments.CourseID = Courses.CourseID;
```

2. Display all enrollments

```
SELECT * FROM Enrollments;
```

Part E

1. What is a Primary Key?

A Primary Key is a column that uniquely identifies each record in a table.

It cannot contain duplicate or NULL values.

2. What is a Foreign Key?

A Foreign Key is a column that creates a relationship between two tables.

It references the Primary Key of another table.